Informativity and Identifiability for Identification of Networks of Dynamical Systems

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***Abstract*— In this paper, we show how informativity and identifiability for networks of dynamical systems can be investigated using Gröbner bases. We provide a sufficient condition for informativity in terms of positive definiteness of the spectrum of external signals and full generic rank of the transfer function relating the external signals to the inputs of the predictor. Moreover, we show how generic local network identifiability can be investigated by computing the dimension of the fiber associated with the closed loop transfer function from external measurable signals to the measured outputs.**

***Index Terms*— System Identification, Networks, Dynamical Systems, Informativity, Generic local network identifiability.**

I. Introduction

SYSTEM identification for networks of dynamical systems is an active research field with a long history. Some early references are [1]–[3]. Some of the work has been focusing on identifying the full network, e.g. [3]–[5], while other work has been focusing on identifying parts of the network, a so-called *sub-network*, e.g. [2], [6]–[11]. The latter is an example of system identification under feedback, a topic that also has a long history, see e.g. [12].

Many system identification methods are based on a so-called *predictor*, and the system identification methods using predictors are called *prediction error methods*. Two fundamental questions related to these methods are what is called *informativity* and *identifiability*. The first question is about if the signals used for system identification are such that the predictor can be uniquely determined. The second question is whether the network topology is such that the open loop transfer functions can be uniquely determined from the predictor. This paper will focus on these questions under the assumption that a stable predictor exists. We do not care if this predictor is linear or nonlinear in the open loop transfer functions. We will investigate these questions both for the full network case and for the sub-network case. We will see that we can handle more general network configurations than what has been possible hitherto in the literature. Specifically, we will consider the *partial measurement case*, i.e. not all of the node signals in the (sub)-network are in the measured outputs. Moreover, we will allow the measurement equation to contain transfer functions to be identified, and we will also allow for direct feed-through from the inputs to the measured outputs. In a seamless way, we also allow for some transfer functions to be known and for rational constraints relating transfer functions to one another.

Regarding informativity, we focus on sufficient conditions in terms of positive definiteness of the spectrum of external signals and full generic rank of the transfer function relating the external signals to the inputs of the predictor. This has previously been studied in e.g. [2], [13]. We specifically generalize this work to the partial measurement case, and to some of the transfer functions to be known. We give both a graph-theoretic and an algebraic geometry characterization of the generic rank.

Moreover, we will derive conditions for generic local network identifiability in terms of dimensions of affine varieties. This is a generalization of the work in [3], [14] and complements the work in [15]. Specifically, we are able to consider some of the transfer functions to be known, and constraints on transfer functions. We are also able to allow for the measurement equation to contain transfer functions to be identified.

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II. Model of Network

We consider a linear dynamic network model defined by

$$\begin{aligned} P(q)w_k &= Q(q)u_k \\ z_k &= R(q)w_k + S(q)u_k \end{aligned} \tag{1}$$

where the quadruple (P, Q, R, S) are transfer function matrices in the forward shift operator q. The dimensions of these transfer function matrices are $n \times n$, $n \times m$, $p \times n$, and $p \times m$, respectively. This also defines the dimensions of all the signals, where u_k is referred to as the input signal, z_k as the output signal, and w_k as the node signal. We assume that P is invertible for almost all q such that the closed loop system is *well-posed*. We may sometimes replace q with the complex variable z. We assume that some of the elements of the transfer functions we have defined are dependent on some parameter θ, which is going to be identified. If we want to emphasize the parameter dependence we may e.g. write $P(q;\theta)$ for transfer functions and $y_k(\theta)$ for signals. We will often drop parameter dependence and dependence on q.

III. Innovation form and Predictor

We divide u_k in a part r_k that we know or measure, and another part e_k, which we refer to as noise. We write

$u_k = (r_k, e_k)$, where r_k has m_r components and e_k has m_e components. We also partition $Q(q)$ and $S(q)$ conformably as

$$Q(q) = \begin{bmatrix} Q_r(q) & Q_e(q) \end{bmatrix}, \quad S(q) = \begin{bmatrix} S_r(q) & S_e(q) \end{bmatrix}.$$

We assume that z_k can be partioned as $z_k = (y_k, r_k)$, where all of z_k is a signal that we have full information about. Here y_k has p_y components, and r_k has p_r components. We conformally write

$$\begin{aligned} \begin{bmatrix} R(q) & S(q) \end{bmatrix} &= \begin{bmatrix} R(q) & S_r(q) & S_e(q) \end{bmatrix} \\ &= \begin{bmatrix} R_y(q) & S_{yr}(q) & S_{ye}(q) \\ 0 & I & 0 \end{bmatrix} \end{aligned}$$

for appropriately defined transfer function matrices. From this, we may conclude that the *closed loop system* from u_k to y_k may be written

$$y_k = G_c(q) r_k + H_c(q) e_k \tag{2}$$

where

$$\begin{aligned} \begin{bmatrix} G_c(q) & H_c(q) \end{bmatrix} &= R_y(q) P(q)^{-1} \begin{bmatrix} Q_r(q) & Q_e(q) \end{bmatrix} \\ &\quad + \begin{bmatrix} S_r(q) & S_e q \end{bmatrix}. \end{aligned} \tag{3}$$

System identification is very often based on a *predictor*, and to this end we define the so-called *innovation form*, i.e. we write

$$\hat{y}_k = G_c(q) r_k + G_o(q) \epsilon_k \tag{4}$$

$$y_k = \hat{y}_k + \epsilon_k \tag{5}$$

for some transfer function matrix $G_o(q)$, where the *innovations* ϵ_k are zero mean equally distributed and independent random variables. We remark that obtaining the innovation form is by no means trivial and often involves solving an algebraic Riccati equation, e.g. [16]. However, we will only need the existence of the innovation form in this work, and no explicit knowledge of the transfer function $G_o(q)$ is needed.

From the above equations we realize that we may write

$$\hat{y}_k = W(q) z_k \tag{6}$$

where

$$W(q) = \begin{bmatrix} W_y(q) & W_r(q) \end{bmatrix} = (I + G_o(q))^{-1} \begin{bmatrix} G_o(q) & G_c(q) \end{bmatrix}.$$

This is called the *predictor* for y_k. Here we need to make the assumption that $W(q)$ has all poles strictly inside the unit circle. This is not a priori given, and has to be verified for each given example. It follows using simple manipulations of the above equations that $G_c(q)$ can be recovered from the predictor as

$$G_c(q) = (I - W_y(q))^{-1} W_r(q). \tag{7}$$

We will now revisit some special cases of the above network formulation that are popular in the literature. Most existing work assume that $P(q) = I - G(q)$ for some transfer function matrix $G(q)$ which is assumed to have a zero diagonal. Most existing work also only consider $S(q) = 0$. Moreover, it is common to have $R_y(q) = I$ implying $y_k = w_k$, which is referred to as the *full measurement* case. The *partial measurement* case is when $R_y(q)$ is a matrix having rows equal to standard basis vectors, which means that y_k contains components of w_k. It is also very common that $Q_r(q)$ and $Q_e(q)$ are diagonal matrices. Moreover, it is not uncommon that they are independent of q. This specifically goes for $Q_r(q)$. Notice that we do not make any assumptions similar to the special cases above.

IV. Informativity

We will now investigate what is called informativity for estimation of the parameter θ. To this end, the following definition is useful:

Definition 1: The signal z_k is said to be *informative enough* if for any two parameters θ_1 and θ_2 such that the closed loop system is stable, it holds that

$$\mathbb{E}\left[\left(\hat{y}_k(\theta_1) - \hat{y}_k(\theta_2)\right) \left(\hat{y}_k(\theta_1) - \hat{y}_k(\theta_2)\right)^T \right] = 0 \tag{8}$$

implies that $W\left(e^{i\omega}; \theta_1\right) = W\left(e^{i\omega}; \theta_2\right)$ for almost all ω.

It holds that

$$\hat{y}(\theta_2)_k - \hat{y}(\theta_1)_k = \Delta W(q) z_k$$

where $\Delta W(q) = W(q; \theta_1) - W(q; \theta_2)$. From Parseval's formula, it follows that (8) is equivalent to

$$\Delta W\left(e^{i\omega}\right) \Phi_z(\omega) \Delta W\left(e^{-i\omega}\right)^T = 0$$

where Φ_z is the spectrum of z_k. Hence, a sufficient condition for the above equality to imply that $W\left(e^{i\omega}; \theta_1\right) = W\left(e^{i\omega}; \theta_2\right)$ for almost all ω, is that $\Phi_z(\omega)$ is positive definite for almost all ω.

We have that the closed-loop transfer function from $u_k = (r_k, e_k)$ to z_k is given by

$$\Pi(q) = R(q) P(q)^{-1} Q(q) + S(q)$$

and that the spectrum of z_k is positive definite if $\Pi\left(e^{i\omega}; \theta_0\right)$ is full row rank for almost all ω and the spectrum of u_k is positive definite. This condition depends on the true system described by θ_0, and hence it is a difficult condition to verify. Let

$$M(q) = \begin{bmatrix} P(q) & Q(q) \\ -R(q) & S(q) \end{bmatrix}. \tag{9}$$

Similarly as in [13], we will instead only investigate the so-called *generic* rank defined as

$$\operatorname{rank}_g \Pi = \max_M \operatorname{rank} \Pi.$$

Here we consider Π to just be a function of the free entries of the matrix M.[1] The only property we need regarding these free entries when we investigate generic rank is that they are algebraically independent[2] commuting indeterminates over the coefficient field. The coefficient field can be taken as the field of real numbers if no transfer functions are known. In case some transfer functions are known, we take the coefficient field as the field of rational functions of transfer functions. This is a field with the standard definition of addition and multiplication. We will later on see that the generic rank is

[1] When no ambiguity arises we will often just refer to M even if we mean the free entries of M.

[2] Algebraically independent is the formal way of saying that the entries are free.

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equal to the rank for almost all M. Since Π is a rational function in z, it follows that

$$\operatorname{rank}_g \Pi(z) = \max_z \operatorname{rank}_g \Pi(z)$$

for almost all $z \in \mathbb{C}$, and hence

$$\operatorname{rank} \Pi(z) = \max_z \operatorname{rank}_g \Pi(z)$$

for almost all M and z. Using this result, it is possible to give a sufficient condition for informativity.

Theorem 1: Assume that the spectrum of $u_k = (r_k, e_k)$ is positive definite and that the generic rank of Π is full, then z_k is informative enough.

Remark 1: If we do not want to use all of u_k for excitation, i.e. if we have some of the components equal to zero, then we just erase the corresponding columns in $M(q)$. Similarly, we might not always be interested in having uniqueness of all of $W(q)$, i.e. we may only be interested in some of the columns. This is easily accommodated by just removing the corresponding rows in $M(q)$.

What has been presented above is a generalization of the results in [13] for a much more general network configuration. We will now give conditions on the generic rank of Π in terms of both a graph and in terms of so-called *Gröbner basis*.

A. Graphical Condition

Since we are investigating generic properties, we drop the dependence on the forward shift operator q or the complex variable z. The following equality is easily proven by multiplying together the matrices:

$$\begin{bmatrix} I & 0 \\ RP^{-1} & I \end{bmatrix} M \begin{bmatrix} I & -P^{-1}Q \\ 0 & I \end{bmatrix} = \begin{bmatrix} P & 0 \\ 0 & RP^{-1}Q + S \end{bmatrix}.$$

From well-posedness, it holds that P is invertible. Hence, the rank of M is the same as the rank of Π plus the dimension of P, which is n. This is a generalization of the formula in (5) in [17] to the direct term case. In that paper, it is shown for real-valued structured[3] matrices (A, B, C) and $s \in \mathbb{C}$, how the generic rank of the transfer function $C(sI - A)^{-1}B$ can be given a graph-theoretic characterization. From algebraic geometry, e.g. [18], it follows that the proof of Theorem 1 in [17] still holds true in our setting, i.e.

$$\{M \mid \operatorname{rank} \Pi < \operatorname{rank}_g \Pi\}$$

is a proper affine variety in the set of all M. Hence it holds that $\operatorname{rank}_g \Pi = \operatorname{rank} \Pi$ for almost all M. Then it is straightforward to see that Theorem 2 of that paper can also be generalized. We are therefore able to derive a graph-theoretic characterization of the generic rank of Π.

To this end, let us define a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, where $\mathcal{V} = \mathcal{U} \cup \mathcal{W} \cup \mathcal{Z}$ with $\mathcal{U} = \{\mathcal{U}_1, \ldots, \mathcal{U}_m\}$, $\mathcal{W} = \{\mathcal{W}_1, \ldots, \mathcal{W}_n\}$, $\mathcal{Z} = \{\mathcal{Z}_1, \ldots, \mathcal{Z}_p\}$. We can now associate each row and column in M with elements in $\mathcal{V}$. The first n columns are labeled $\mathcal{W}_1$ to $\mathcal{W}_n$, and the next m columns $\mathcal{U}_1$ to $\mathcal{U}_m$. We proceed similarly with the rows. We then define the edges in $\mathcal{E}$ as all the ordered two-tuples (v_i, v_j) with $v_i, v_j \in V$ such that M_{v_j, v_i} is not a structural zero. We do however, not include any self-loop edges, i.e. edges for which $v_i = v_j$. If there are vertices $v_1, \ldots, v_k$ in $\mathcal{V}$ such that $(v_i, v_{i+1}) \in \mathcal{E}$ for $i = 1, \ldots, k-1$, we say that there is a path in $\mathcal{G}$ from v_1 to v_k. If $v_1 \in \mathcal{U}$ and $v_k \in \mathcal{Z}$ we say that there is a path from $\mathcal{U}$ to $\mathcal{Z}$. We say that two paths from $\mathcal{U}$ to $\mathcal{Z}$ are *disjoint* if they have no vertex in common. We say that an l-tuple of paths from $\mathcal{U}$ to $\mathcal{Z}$ is disjoint if each pair of paths in the l-tuple is disjoint.

We are now able to state the following result, which is a generalization of Theorem 2 in [17].

Theorem 2: The maximum number of disjoint paths in $\mathcal{V}$ from $\mathcal{U}$ to $\mathcal{Z}$ is equal to the generic rank of Π.

Remark 2: The problem of finding the maximum number of disjoint paths can be formulated as a maximum flow problem for an associated flow graph.

Remark 3: For the above proof to hold, we need to assume that all non-zero entries of M are free variables. This assumption is not needed for the Gröbner basis condition to follow.

B. Gröbner Basis Condition

We will now investigate the generic rank of Π using Gröbner basis. We are still considering Π to be a rational function of the free entries of M. We let D_i be the least common multiple of the denominators of the ith row of Π. Multiply each row of Π with this polynomial to obtain the polynomial matrix $\tilde{\Pi}$. Then the generic rank of Π, when well-posedness holds, is the same as the generic rank of $\tilde{\Pi}$ when $D_i \neq 0$. Now, define the ideal $\mathcal{I}_k$ generated by all the minors of $\tilde{\Pi}$ of dimension k together with the polynomial $1 - tD$, where D is the product of all D_i and t is an additional variable. Notice that the equation $1 - tD = 0$ always has a solution where $D \neq 0$ and $t = 1/D$.[4] From this the following theorem follows.

Theorem 3: Given the ideal $\mathcal{I}_k$ defined above it holds:

1) If the reduced Gröbner basis of $\mathcal{I}_k$ is one, then $\operatorname{rank} \Pi \geq k$.
2) If the reduced Gröbner basis of $\mathcal{I}_k$, excluding the basis depending on t, is zero, then $\operatorname{rank} \Pi < k$.
3) Otherwise $\operatorname{rank}_g \Pi \geq k$. Then the Gröbner basis, excluding the basis depending on t, defines the set of entries M for which $\operatorname{rank} \Pi < \operatorname{rank}_g \Pi$.

Remark 4: The lexicographic ordering should be used when the Gröbner basis is computed, and the variable t should be the largest variable according to that ordering.

Remark 5: In case some of the entries of M are known, then one can instead of polynomials with coefficients in the field $\mathbb{R}$ consider polynomials with coefficients in the field of rational functions of transfer functions in the known entries of M. This will, in most cases, speed up the computation of the Gröbner basis.

Remark 6: In case some of the entries of M are constrained by any rational equation, then one can just add this to the ideal.

[3]Notice that we also have structured matrices, since the transfer function matrices are often sparse.

[4]Notice that $\det P$ might have zeros that are not zeros of D, and hence the graph results might be conservative as compared to the Gröbner basis results.

Specifically, one may consider constraining some entries to be the same. However, then it is easier to just remove the duplicate variable to start with.

Notice that the generalizations in the remarks above are trivial to include in the Gröbner basis condition. This is not so easy to do in the graphical condition of the previous subsection, which assumes all non-zero transfer functions to be free variables.

V. Identifiability

As was discussed in Section III the closed loop transfer function can be recovered from the predictor using (7), which we can uniquely do under the assumption of informativity. The next question that arises is whether we can recover parts of M from G_c. From (3), it holds that

$$G_c = R_y P^{-1} Q_r + S_{yr}$$

which is a rational expression in the free entries of (P, Q_r, R_y, S_{yr}), which we will denote by the k-dimensional vector X. Hence, a relevant question is when a rational equation has a unique solution. However, in many cases, this is too much to ask for. What is a more relevant question is whether there is a finite number of isolated solutions for a given G_c. To investigate this question, let F be a function defined by the rule

$$F(X) = R_y P^{-1} Q_r + S_{yr}$$

Similarly as in [15], we consider the domain V_o of F to be the set of all complex-valued vectors X of dimension k for which the inverse of P exists. This is a dense open subset of $\mathbb{C}^k$. We then define the set of *admissible* G_c as $V_c = F(V_o)$. If we take V_c to be the co-domain of F, it follows that F is a surjective function.

Unique solution and finite number of isolated solutions parallels the discussion in [19] regarding *generic network identifiability* and *generic local network identifiability*, where it is argued that the latter is easier to investigate than the former. There it is also conjectured that generic local network identifiability implies generic network identifiability.

Definition 2: We say that we have generic local network identifiability if for almost all $G_c \in V_c$ the following hold. Given a solution $X \in V_o$ to

$$F(X) = G_c \tag{10}$$

there exists $\epsilon > 0$ such that for any $\tilde{X}$ satisfying $\|\tilde{X} - X\| < \epsilon$ and $F(X) = F(\tilde{X})$ it follows that $\tilde{X} = X$.

Remark 7: From this definition it can be shown that if a network is generically locally identifiable, then it can be recovered up to a discrete ambiguity, i.e. the set of X corresponding to the same G_c is discrete except for a set of G_c of measure zero, [19].

Then we have the following theorem.

Theorem 4: It holds that X is generically locally network identifiable for any admissible G_c from (10) if and only if

$$\dim(V_o) = \dim(V_c).$$

Proof: For a fixed $G_c \in V_c$ define the fiber

$$F^{-1}(G_c) = \{X \in V_o \mid F(X) = G_c\}.$$

It holds that generic local network identifiability is equivalent to that $\dim\left(F^{-1}(G_c)\right) = 0$ for almost all $G_c \in V_c$. Since F is surjective, it follows from the fiber dimension theorem in algebraic geometry that

$$\dim(V_o) = \dim(V_c) + \dim\left(F^{-1}(G_c)\right)$$

for almost all $G_c \in V_c$. The result is immediate from this. ■

Remark 8: Since V_o is a dense subset of $\mathbb{C}^k$ the dimension of V_o is k. We define the polynomial matrices S and T such that $T \odot F(X) = S$ where $\odot$ is the Hadamard product of matrices. We let D be the least common multiple of the entries of T. Then we define the polynomial matrix $T \odot G_c - S$, and the ideal generated by the entries of this matrix and $1 - tD$. We then compute the Gröbner basis for this ideal using the lexicographic ordering, where the variables in G_c should be the smallest variables according to this ordering. The part of the basis containing only the variables in G_c defines the smallest variety that contains V_c, c.f. [20, Ch.3.3]. The dimension of this variety is the same as the dimension of V_c. Then the algorithm in [21] can be used to compute the dimension.

Remark 9: In case some of the entries of X are known, we just exclude them when defining V_o and the variable X. When computing the Gröbner basis, one then considers rational functions with coefficients in the field of rational functions of the transfer functions in the known entries of X rather than coefficients in the field $\mathbb{R}$.

Remark 10: It is possible to put rational constraints on X. In case the constraint is that one of the entries is the same as another entry, then it is easier to just replace one variable with the other to remove the constraint We will discuss this in an example later on in more detail.

Remark 11: Notice that it is possible to discuss identifiability also when not all of the predictor is estimated uniquely. Then we just consider the closed loop transfer function corresponding to the part of the predictor we estimate uniquely. Then there will be fewer equations that determine the open loop transfer functions, but there could still be enough equations to guarantee generic network identifiability. We will discuss this in more detail for an example later on.

What has been presented above is a generalization of the results in [3] to the case when $Q_r \neq I$. It is also a generalization of the results in [14] to the case when $R_y \neq I$. Our result also generalizes the results in [15], where it is assumed that Q_r and R_y are given zero-one matrices. We also consider a direct term S_{yr}, which other work do not.

VI. Sub-Networks

So far we have discussed the identification of a whole network. Often one is only interested in parts of a network. Without loss of generality we will assume that the part of the network we are interested in corresponds to the leading entries w_k^A of $w_k = (w_k^A, w_k^B)$. We then partition M as

$$M = \begin{bmatrix} P & Q_r & Q_e \\ R_y & S_{yr} & S_{ye} \\ 0 & I & 0 \end{bmatrix} = \begin{bmatrix} P_A & P_{AB} & Q_{Ar} & Q_{Ae} \\ P_{BA} & P_B & Q_{Br} & Q_{Be} \\ R_{yA} & R_{yB} & S_{yr} & S_{ye} \\ 0 & 0 & I & 0 \end{bmatrix}.$$

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We have

$$P_A w_k^A = \begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix} \begin{bmatrix} w_k^B \\ r_k \end{bmatrix} + Q_{Ae} e_k$$
$$y_k = R_{yA} w_k^A + \begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix} \begin{bmatrix} w_k^B \\ r_k \end{bmatrix} + S_{ye} e_k.$$

Hence, we can investigate generic local network identifiability of network A by making the following replacements in our previous definitions:

$$\begin{aligned} P_A &\to P \\ \begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix} &\to Q_{Ar} \\ R_{yA} &\to R_y \\ \begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix} &\to S_{yr} \\ w_k^A &\to w_k \\ (w_k^B, r_k) &\to r_k. \end{aligned}$$

We then obtain the following rational equation for investigating identifiability:

$$R_{yA} P_A^{-1} \begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix} + \begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix} = G_c.$$

If there are columns in

$$\begin{bmatrix} -P_{AB} & Q_{Ar} \\ R_{yB} & S_{yr} \end{bmatrix}$$

that are zero, then they may be removed from the rational function. Similarly any zero rows in $\begin{bmatrix} R_{yA} & S_{yr} \end{bmatrix}$ may be removed. When it comes to informativity, we realize that the B-part of the network provides excitation of the A-part of the network, and hence the whole network should be considered. Since w_k^B is part of the exogenous input, we should consider

$$M = \begin{bmatrix} P_A & P_{AB} & Q_{Ar} & Q_{Ae} \\ P_{BA} & P_B & Q_{Br} & Q_{Be} \\ R_{yA} & R_{yB} & S_{yr} & S_{ye} \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix}$$

when we investigate informativity. However, rows of the second last block row that correspond to the zero columns in $\begin{bmatrix} -P_{AB} \\ R_{yB} \end{bmatrix}$ discussed above should be removed.

As seen above, the exogenous signal contains also measurements of the node signals in the B-part of the network. We also need to estimate models for parts of the B-network. Because of sparsity this might not be a big problem, since we may not have to measure all of w_k^B. However, this is in many cases still not desirable. If we instead of measuring the node signals w_k^B can measure

$$\begin{bmatrix} \tilde{w}_k^B \\ \hat{w}_k^B \end{bmatrix} = \begin{bmatrix} -P_{AB} \\ R_{yB} \end{bmatrix} w_k^B,$$

we will not need to estimate models for any parts of the B-network. Then we obtain the equations:

$$P_A w_k^A = \begin{bmatrix} I & 0 & Q_{Ar} \end{bmatrix} \begin{bmatrix} \tilde{w}_k^B \\ \hat{w}_k^B \\ r_k \end{bmatrix} + Q_{Ae} e_k$$
$$y_k = R_{yA} w_k^A + \begin{bmatrix} 0 & I & S_{yr} \end{bmatrix} \begin{bmatrix} \tilde{w}_k^B \\ \hat{w}_k^B \\ r_k \end{bmatrix} + S_{ye} e_k.$$

The replacement will instead be:

$$\begin{aligned} P_A &\to P \\ \begin{bmatrix} I & 0 & Q_{Ar} \end{bmatrix} &\to Q_r \\ Q_{Ae} &\to Q_e \\ R_{yA} &\to R_y \\ \begin{bmatrix} 0 & I & S_{yr} \end{bmatrix} &\to S_{yr} \\ S_{ye} &\to S_{ye} \\ w_k^A &\to w_k \\ (\tilde{w}_k^B, \hat{w}_k^B r_k) &\to r_k. \end{aligned}$$

We then obtain the following rational equation for investigating identifiability:

$$R_{yA} P_A^{-1} \begin{bmatrix} I & 0 & Q_{Ar} \end{bmatrix} + \begin{bmatrix} 0 & I & S_{yr} \end{bmatrix} = G_c$$

which now contains dynamics of only the A-part of the network. Any zero rows in $\begin{bmatrix} Ry_A & S_{yr} \end{bmatrix}$ may be removed. If there are zero rows in

$$\begin{bmatrix} -P_{AB} \\ R_{yB} \end{bmatrix}$$

it follows that there are zero signals in $(\tilde{w}_k^B, \hat{w}_k^B)$. The corresponding columns in the above rational matrix may then be removed. Also any zero columns in $\begin{bmatrix} Q_{Ar} \\ S_{yr} \end{bmatrix}$ may be removed. Since $(\tilde{w}_k^B, \hat{w}_k^B)$ is part of the exogenous input, we should consider

$$M = \begin{bmatrix} P_A & P_{AB} & Q_{Ar} & Q_{Ae} \\ P_{BA} & P_B & Q_{Br} & Q_{Be} \\ R_{yA} & R_{yB} & S_{yr} & S_{ye} \\ 0 & -P_{AB} & 0 & 0 \\ 0 & R_{yB} & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix}$$

when we investigate informativity. However, any zero rows in the fourth and fifth block rows should be removed, since they correspond to zero signals in $(\tilde{w}_k^B, \hat{w}_k^B)$.

Notice that the criterion for informativity holds for any pattern of the matrices defining the M-matrix. Based on this, we now realize that we are able to analyze informativity and generic local network identifiability for sub-networks using the methods we have developed for a whole network. It is possible to combine the two above methods depending on what can be measured so that only some of the columns of P_{AB} and R_{yB} are estimated.

VII. Example

In this section, we will investigate the theory we have developed on some examples.

Example 1: We will look at the very simple example in Figure 1 to illustrate why the conditions we have presented are more general than the ones previously presented in the literature. We assume that $v_k^1 = H^1 e_k^1$, $v_k^2 = H^2 e_k^2$ and that $r_k^1 = Q^1 r_k$ and $r_k^2 = Q^2 r_k$. Notice that we have the same single r_k entering the loop in two places. We are interested in estimating (G^1, Q^1), and use a predictor for predicting w_k^1. We can measure (w_k^3, r_k) in addition to w_k^1, and therefore the

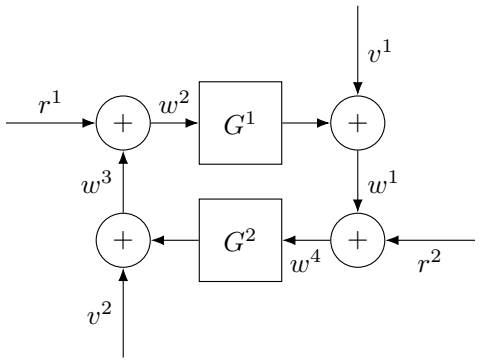


Fig. 1. Block diagram for a simple network.

predictor input is $z_k = (w_k^1, w_k^3, r_k)$. We obtain the following equations

$$\begin{bmatrix} w_k^1 \\ w_k^2 \\ w_k^3 \\ w_k^4 \end{bmatrix} = \begin{bmatrix} 0 & G^1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & G^2 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} w_k^1 \\ w_k^2 \\ w_k^3 \\ w_k^4 \end{bmatrix} + \begin{bmatrix} 0 & H^1 & 0 \\ Q^1 & 0 & 0 \\ 0 & 0 & H^2 \\ Q^2 & 0 & 0 \end{bmatrix} \begin{bmatrix} r_k \\ e_k^1 \\ e_k^2 \end{bmatrix}$$
$$z_k = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} w_k^1 \\ w_k^2 \\ w_k^3 \\ w_k^4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r_k. \tag{11}$$

It is straightforward to see that there are three disjoint paths: $(\mathcal{E}_1, \mathcal{W}_1, \mathcal{Z}_1)$, $(\mathcal{E}_2, \mathcal{W}_3, \mathcal{Z}_2)$ and $(\mathcal{R}_1, \mathcal{Z}_3)$, and hence we have informativity, assuming that the spectrum for (r_k, e_k^1, e_k^2) is positive definite. We can also verify this by directly looking at the transfer function

$$\Pi = \begin{bmatrix} \frac{G^1(Q^1+G^2Q^2)}{1-G^1G^2} & \frac{1}{1-G^1G^2} & \frac{G^1}{1-G^1G^2} \\ \frac{G^2(G^2Q^1+Q^2)}{1-G^1G^2} & \frac{G_2}{1-G^1G^2} & \frac{1}{1-G^1G^2} \\ 1 & 0 & 0 \end{bmatrix}$$

which has determinant $-1/(1-G^1G^2)$. Hence, both the rank and the generic rank are full when we have well-posedness.

The closed loop system transfer function from (w_k^3, r_k) to w_k^1 is given by

$$\begin{bmatrix} G_c^1 & G_c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -G^1 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 & 0 \\ 1 & Q^1 \end{bmatrix} = \begin{bmatrix} G^1 & G^1Q^1 \end{bmatrix}.$$

The dimension of the affine varieties are $\dim V_c = \dim V_o = 2$ and we conclude that

$$(G^1, Q^1) = (G_c^1, G_c^2/G_c^1).$$

Hence, we have generic local network identifiability. The reason we do not have network identifiability is that there only exists a unique solution for all G_c^2 when $G_c^1 \neq 0$. Under this assumption, there is only one solution for each given G_c, and hence we do not only have generic local network identifiability, but generic network identifiability.

Example 2: Let us now consider the case where we are only interested in estimating G^1, and where we do not care about any of the other transfer functions. We also assume that Q^1 is known. We still use the same measured signals and the same predictor. The closed system transfer functions are the same. Now, the equations for generic network identifiability are linear in G^1 since Q^1 is no longer a variable but known. There are two equations, but only one unknown. Because of this, we have more information than what is needed. We can therefore, if we so desire, consider to only estimate parts for the predictor, either the part corresponding to w_k^3 or to r_k. Hence, we only need informativity for one of these signals. Therefore it is enough if either the spectrum for (r_k, e_k^1) or the spectrum for (e_k^1, e_k^2) is positive definite. This shows an interesting interplay between informativity and generic local network identifiability. For more complicated networks, it is therefore not so easy to know what signals should be excited to be able to obtain consistent estimates.

Example 3: We consider again the network in Figure 1, but this time we are interested in estimating both transfer functions (G^1, G^2). We perform two separate experiments where we first excite with only r_k^1 and measure w_k^1, and then we excite with only r_k^2 and measure w_k^3. It is not difficult to conclude that we have informativity for both experiments, irrespective of what the noises are. It is also straightforward to conclude that

$$w_k^1 = G_c^1 r_k^1, \quad w_k^3 = G_c^2 r_k^2$$

where

$$G_c^1 = \frac{G^1}{1-G^1G^2}, \quad G_c^2 = \frac{G^2}{1-G^1G^2}.$$

If we replace (G^1, G^2) with $(-1/G^2, -1/G^1)$ we obtain the same (G_c^1, G_c^2). Hence, this is an example where local identifiability does not imply identifiability. Notice that

$$\begin{bmatrix} w_k^1 \\ w_k^e \\ w_k^3 \\ w_k^4 \end{bmatrix} = \frac{1}{1-G^1G^2} \begin{bmatrix} G^1 & G^1G^2 \\ 1 & G^2 \\ G^1G^2 & G^2 \\ G^1 & 1 \end{bmatrix} \begin{bmatrix} r_k^1 \\ r_k^2 \end{bmatrix}$$

and it is not possible to pick out these two closed loop transfer functions by multiplying from the left and the right with zero-one matrices. However, if we duplicate the network and constrain the transfer functions in the two networks to be the same, it is possible to pick out them with zero-one matrices. This shows that the conjecture in [19] that local identifiability implies identifiability is not necessarily true if the same transfer functions appear twice in a network.

VIII. Numerical Examples

The above examples were possible to analyze without using any of the theory we have developed. In this section, we will show how a Matlab implementation based on Gröbner basis calculations can be used to analyze fairly large examples, that are not so easy to analyze by hand.

Example 4: Consider the dynamical network defined by

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(P, Q_r, R_y, S_{yr}), where $P = I - G$, $S_{yr} = 0$ and

$$G = \begin{bmatrix} 0 & G^{12} & 0 & 0 & 0 \\ 0 & 0 & G^{23} & 0 & 0 \\ 0 & 0 & 0 & G^{34} & 0 \\ 0 & 0 & 0 & 0 & G^{45} \\ G^{51} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_r = \begin{bmatrix} Q^1 & 1 & 0 & 0 & 0 \\ 0 & Q^2 & 1 & 0 & 0 \\ 0 & 0 & Q^3 & 1 & 0 \\ 0 & 0 & 0 & Q^4 & 1 \\ Q^5 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_y = \begin{bmatrix} R^1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

For this example, we have $\dim V_o = 11$, and using Matlab, we can compute $\dim V_c = 11$ in 11 minutes on a laptop computer. Hence, we have generic local network identifiability. In a second we can verify that the row rank of Π is full for the case when $Q_e = I$ and $S_{ye} = 0$, and therefore we also have informativity if the spectrum of (r_k, e_k) is positive definite.

Example 5: We consider the same dynamic network as in the previous example, but now we are only interested in identifying the transfer functions (G^{12}, Q^2, Q^2, R^1). We let $y_k = (w_k^1, w_k^2)$ and we measure $\tilde{w}_k^B = G^{23} w_k^3$ from the B-part of the network. We then have that

$$F(X) = \begin{bmatrix} R^1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -G^{12} \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 & Q^1 & 1 & 0 \\ 1 & 0 & Q^2 & 1 \end{bmatrix}$$

with the input $(\tilde{w}_k^B, r_k^1, r_k^2, r_k^3)$ and the output y_k. The M-matrix for investigating informativity is

$$M = \begin{bmatrix} P & Q_r & Q_e \\ R_y & S_{yr} & S_{ye} \\ \tilde{P} & 0 & 0 \\ 0 & I & 0 \end{bmatrix}$$

where P and Q_r are as in the previous example, $Q_e = I$, $S_{yr} = 0$, $S_{ye} = 0$,and

$$R_y = \begin{bmatrix} R^1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad \tilde{P} = \begin{bmatrix} 0 & 0 & G^{23} & 0 & 0 \end{bmatrix}.$$

Notice that the zero matrices have one fewer row as compared to the previous example. In about a second we can verify that we have generic local network identifiability and informativity. We can also investigate if (r_k^4, r_k^5) can be equal to zero. We just put the corresponding rows and columns equal to zero in M, and we obtain that we still have informativity. It is also possible to investigate if we in addition can have r_k^3 equal to zero. However, then we have to investigate generic local network identifiability for the case when we remove the last column in $F(X)$. It turns out that we have both informativity and generic local network identifiability also for this case.

IX. Conclusions

In this paper, we have shown how to investigate informativity and generic local network identifiability for networks of dynamical systems. Our informativity results are characterized in terms of vertex-disjoint paths of a graph associated with the dynamic network. Informativity can also be cast as an algebraic geometry problem where Gröbner bases are computed. The generic local network identifiability result is phrased as two affine varieties having the same dimension. This can also be investigated using Gröbner bases. Our approach seamlessly generalizes to the case where some transfer functions are known. This is in contrast to the work presented in e.g. [14]. Actually, our framework takes advantage of known transfer functions in the computations of the Gröbner basis in that computation time will be lower. Future work will focus on investigating the interplay between informativity and generic local network identifiability in more detail.

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